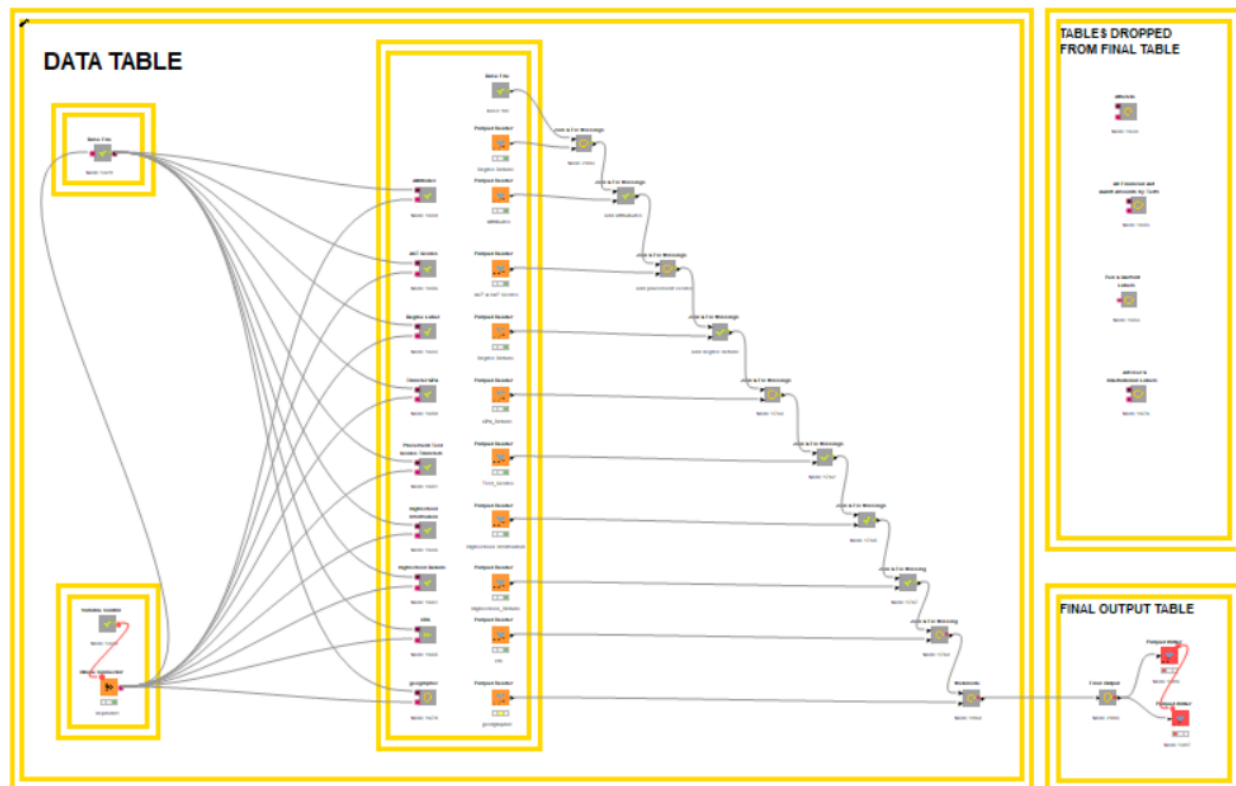


DATA TABLE WRANGLING OF UIC STUDENT PREDICTION



To obtain all the required attributes from the table, the data wrangling process was performed within the KNIME platform. In the lower-left corner, you will find the connection to the Oracle database, specifically DSPROD01. The preceding node above the connection serves as a flow variable, allowing for control over the time range from which the data was extracted. The data encompasses snapshots of term codes starting from fall 2013 up to the latest summer of 2023. These term codes are utilized as flow variables and are incorporated into all subsequent queries.

The workflow begins by establishing a database connection and enabling the passage of flow variables throughout the rest of the workflow. Next, the base file table is created. All data is then left-joined to this table to incorporate additional attributes. The light grey nodes represent metanodes, each containing a loaded and transformed table to be joined with the base file. All nodes export the data as parquet files, which are subsequently read back into the workflow and joined with the base file. Each downward step adds another table to the process. The diagonal line of light grey nodes includes the necessary transformations before and after the join, along

with a preliminary process to handle missing values at each step. Once all the tables are joined, the final output is exported as a parquet file to the right.

Another section includes tables that were initially considered for inclusion but were later dropped. These tables are included here for ease of joining in case they are decided to be used in the future. The athlete table and adviser information were dropped due to a high occurrence of missing values. The pell & stafford loan table, as well as financial aid information, were also dropped as they were not permitted for use. However, their potential usage is currently under review for future consideration.

Student Base File

```
SELECT
  a.EDW_PERS_ID,
  a.REG_SNAPSHOT_KEY,
  a.REG_SNAPSHOT_DESC,
  a.DEPT_NAME,
  a.SEX_CD,
  a.PERS_MARITAL_STATUS_DESC,
  a.PERS_CITZN_TYPE_DESC,
  a.PERS_CITZN_TYPE_GROUP,
  a.RACE_ETH_DESC,
  a.IPEDS_RACE_ETH_DESC,
  a.AGE,
  a.IPEDS_RACE_ETH_CATGRY_DESC,
  a.RACE_ETH_RPT_CD AS IPEDS_RE_CATGRY_CD,
  a.RACE_ETH_RPT_DESC AS IPEDS_RE_CATGRY_DESC,
  a.DEATH_IND,
  b.STUDENT_TYPE_DESC,
  b.STUDENT_TYPE_CD,
  b.EXPCT_GRAD_TERM_CD,
  b.STUDENT_RES_DESC,
  b.STUDENT_RES_GROUP,
  b.AH_ACAD_STD_TERM_CD,
  b.AH_ACAD_STD_DESC,
  b.CALC_CLS_DESC,
  b.STUDENT_EDUC_GOAL_DESC,
  b.ADM_APPL_CURR_1_TYPE_DESC,
  b.STUDENT_ADMIT_TERM_CD,
  b.ADMIN_COLL_NAME,
  b.ACAD_COLL_NAME,
  b.STUDENT_DEPT_NAME,
  b.STUDENT_LEVEL_DESC,
  b.STUDENT_CURR_1_DEG_CD,
```

```

b.STUDENT_CURR_1_DEG_NAME,
b.STUDENT_CURR_1_MAJOR_NAME,
b.LEVEL_CREDIT_GROUP,
b.STUDENT_CURR_1_CATLG_TERM_CD,
b.STUDENT_CURR_1_CATLG_TERM_DESC,
d.TERM_CD,
d.ACAD_YR_CD,
e.STUDENT_HOLD_TYPE_CD,
e.STUDENT_HOLD_TYPE_DESC,
e.STUDENT_HOLD_START_DT,
e.STUDENT_HOLD_END_DT,
e.STUDENT_HOLD_REASON_CD,
e.STUDENT_HOLD_REASON_DESC,
e.STUDENT_HOLD_OWE_AMT,
e.STUDENT_HOLD_ORIG_CD,
e.STUDENT_HOLD_RELEASE_IND,
e.STUDENT_REG_HOLD_IND,
e.STUDENT_TRNSCPT_HOLD_IND,
e.STUDENT_GRAD_HOLD_IND,
e.STUDENT_GRADE_RPT_HOLD_IND,
e.STUDENT_ENRL_VERF_HOLD_IND,
e.STUDENT_APPL_HOLD_IND,
e.STUDENT_COMPL_HOLD_IND,
g.STUDENT_STATUS_CD,
g.STUDENT_STATUS_DESC,
g.EXPCT_GRAD_ACAD_YR_CD,
g.STUDENT_LOA_CD,
g.STUDENT_LOA_DESC,
g.STUDENT_LOA_START_DT,
g.STUDENT_LOA_END_DT
FROM
    EDW.T_RS_PERS a
LEFT JOIN
    EDW.T_RS_STUDENT b ON a.EDW_PERS_ID = b.EDW_PERS_ID
    AND a.REG_SNAPSHOT_KEY = b.REG_SNAPSHOT_KEY
LEFT JOIN
    EDW.T_RS_STUDENT_REG c ON a.EDW_PERS_ID = c.EDW_PERS_ID
    AND a.REG_SNAPSHOT_KEY = c.REG_SNAPSHOT_KEY
LEFT JOIN
    EDW.T_RS_TIME d ON a.REG_SNAPSHOT_KEY = d.REG_SNAPSHOT_KEY
LEFT JOIN
    EDW.T_RS_STUDENT_HOLD e ON a.EDW_PERS_ID = e.EDW_PERS_ID
    AND a.REG_SNAPSHOT_KEY = e.REG_SNAPSHOT_KEY
LEFT JOIN

```

```

EDW.T_STUDENT_TERM g ON a.EDW_PERS_ID = g.EDW_PERS_ID
AND d.TERM_CD = g.TERM_CD
WHERE
a.CAMPUS_CD = '200'
AND b.STUDENT_STATUS_DESC LIKE 'Active%'
AND b.REG_SNAPSHOT_DESC = 'Census'
AND c.STUDENT_REG_IND = 'Y'
AND d.TERM_CD IN ($${Sstr_codes}$)

```

The Base file consists of fundamental student attributes obtained from tables within the EDW schema.

1. EDW.T_RS_PERS: Basic demographic information
2. EDW.T_RS_STUDENT: Basic student information
3. EDW.T_RS_STUDENT_REG: Registration indications
4. EDW.T_RS_TIME: Term Codes
5. EDW.T_RS_STUDENT_HOLD: Holds for registration indicators
6. EDW.T_STUDENT_TERM: Possible leave of absence codes

Degree Table

Note: “\$\${Sstr_codes}\$” is passed from the flow variable creator.

```

SELECT f.EDW_PERS_ID,
f.DEG_SEQ_NBR,
f.STUDENT_AH_DEG_CUR_INFO_IND,
f.GNRL_STUDENT_RECORD_TERM_CD,
f.GNRL_STUDENT_RECORD_TERM_DESC,
f.GRAD_WORK_COMPLETION_TERM_CD,
f.GRAD_WORK_COMPLETION_TERM_DESC,
f.GRAD_TERM_CD,
f.GRAD_TERM_DESC,
f.GRAD_ACAD_YR_CD,
f.GRAD_STATUS_CD,
f.GRAD_STATUS_DESC,
f.DEG_BULL_YR,
f.DEG_STATUS_CD,
f.DEG_STATUS_DESC,
f.DEG_CURR_1_CATLG_TERM_CD
FROM EDW.T_STUDENT_AH_DEG_HIST f
WHERE GRAD_TERM_CD IN ($${Sstr_codes}$)

```

The table in question holds crucial information regarding a student's degree attainment and graduation status. The f.STUDENT_AH_DEG_CUR_INFO_IND was a vital component in labeling students as "graduated" within the target variable.

Attributes Table

```
SELECT
*
FROM edw.t_rs_student_attr
WHERE student_attr_cd in ('2145','2400','2405',
'2411','2412','2413','2040','2045','2050','2060',

'2070','2075','2080','2081','2085','2090','2095','2100','2105','2110','2120','2125','2130','213
5','2081',
'2065','2140', '2002', '2003', '2SES','2SEG')
AND REG_SNAPSHOT_DESC = 'Census'
```

These attributes encompass a range of codes with various possible labels. This table serves as a comprehensive repository for different student designations, such as "first-generation students," "honors students," or students affiliated with special programs that offer unique financial aid benefits. The list of labels is extensive, and for detailed information, please consult the data cookbook. These attributes were included to explore correlations between these labels and student outcomes. However, it was found that only a few of them exhibited significant correlation with the target variable.

SAT & ACT Scores Table

```
select
DISTINCT EDW_PERS_ID,
MAX_ACT_COMPOSITE_SCORE AS act_composite,
MAX_ACT_ENGL_SCORE as act_english,
MAX_ACT_MATH_SCORE as act_math,
MAX_ACT_READING_SCORE as act_reading,
MAX_ACT_SCIENCE_REASON_SCORE as act_science
/*New SAT fields added Fall 2017*/
--MAX_SAT_TOT_SCORE,
--MAX_POST_2016_SAT_MATH_SCORE,
--MAX_SAT_READ_WRITING_SCORE,
--MAX_SAT_WRITING_LANG_SCORE
from edw.T_RS_MAX_TEST_SCORE
```

It is worth mentioning that ACT and SAT scores are no longer mandatory requirements. Consequently, this table contains a significant number of missing values. Despite this, the decision was made to retain the table, as the presence of an SAT or ACT score may still serve as a positive indicator for the continuation of coursework. However, it is important to note that the scores themselves were not included, but rather their presence was utilized as an indicator in the analysis.

Degree History Table

```
SELECT
DEG_CD,
DEG_NAME,
FINANCIAL_AID_COUNT_IND,
DEG_LEVEL_CD,
AWARD_CATGRY_CD
FROM EDW.T_DEG_CD_HIST
where deg_cd_cur_info_ind = 'Y'
```

This table include a few additional features about the students degree history not already included in the base file.

Transfer GPA Details

```
SELECT
*
--c.EDW_PERS_ID,
--c.level_gpa_hour as T_gpa_hours ,
--c.level_gpa_qual_pt as T_gpa_qual,
--c.level_gpa as T_gpa
FROM edw.t_rs_student_ah_level_gpa c
WHERE CAMPUS_CD = '200'
and c.level_gpa_type_ind in ('T','I')
```

This table contains GPA details and transfer information.

Placement Scores

```
--%let math='2P53'
--%let eng ='2P26'
```

```
--%let chem='2P10'
SELECT
*
--EDW_PERS_ID,
--test_cd,
--test_type_score_name,
--test_result_test_score,
--test_result_src_desc
from edw.V_adm_test_result_hist
where test_result_campus_cd = '${SV_campus_cd}$'
and test_result_cur_info_ind= 'Y'
and test_cd in ('${Sv_math}$', '${Sv_eng}$', '${Sv_chem}$')
```

The table comprises scores for placement exams that new students are required to take prior to enrolling in classes. These scores pertain to assessments in Math, English, and Chemistry. It was observed that higher scores on these exams exhibited a positive correlation with both retention and graduation rates.

Highschool Details Table

```
SELECT
a.SBGI_CHARISTIC_DEMO_YR,
a.SBGI_CHARISTIC_CD,
a.SBGI_CHARISTIC_DESC,
a.SBGI_CHARISTIC_EFF_DT,
a.SBGI_CHARISTIC_POST_DT,
b.SBGI_CD,
b.SBGI_TYPE_CD,
b.SBGI_TYPE_DESC,
b.SBGI_RECRUIT_SRC_IND,
b.SBGI_NAME,
b.FICE_CD,
b.SBGI_STREET_LINE_1,
b.SBGI_STREET_LINE_2,
b.SBGI_STREET_LINE_3,
b.SBGI_CITY,
b.SBGI_STATE_CD,
b.SBGI_STATE_NAME,
b.SBGI_COUNTY_CD,
b.SBGI_COUNTY_NAME,
b.SBGI_ZIP_CD,
b.SBGI_NATION_CD,
b.SBGI_NATION_NAME,
```

```

b.SBGI_CD_EFF_DT,
b.SBGI_CD_POST_DT
FROM edw.t_sbgi_charistic a
INNER JOIN edw.t_sbgi_cd b
ON a.SBGI_CD = b.SBGI_CD
WHERE a.SBGI_CHARISTIC_CD = 'P'

```

This table contains specific information about the highschool from which a student has come from.

This table was joined with the table below on SBGI_CD = HS_SBGI_CD

```

SELECT DISTINCT
b.EDW_PERS_ID,
b.TERM_CD,
c.APPL_LEVEL_CD,
c.APPL_STU_TYPE_CD,
d.HS_SBGI_CD,
e.HS_SUBJ_CD,
f.FICE_CD,
f.SBGI_NAME,
f.SBGI_CITY,
f.SBGI_STATE_CD,
f.sbgi_zip_cd,
f.sbgi_county_cd,
f.sbgi_county_name
FROM DM_APPL.V_AC_APPL_CENSUS a
INNER JOIN DM_APPL.V_AC_APPL b
ON a.APPL_KEY = b.APPL_KEY
INNER JOIN DM_APPL.V_AC_APPL_CURR c
ON a.APPL_CURR_KEY = c.APPL_CURR_KEY
INNER JOIN DM_APPL.V_AC_HS d
ON a.APPL_PERS_KEY = d.APPL_PERS_KEY
INNER JOIN DM_APPL.V_AC_HS_SUBJ e
ON d.HS_SBGI_CD = e.SBGI_CD
AND d.APPL_PERS_KEY = e.APPL_PERS_KEY
INNER JOIN EDW.T_SBGI_CD f
ON e.SBGI_CD = f.SBGI_CD
WHERE
--b.TERM_CD= ${Sv_term_cd} AND
c.APPL_LEVEL_CD = '2U' AND
c.APPL_STU_TYPE_CD in ('F','S') AND /*added S for summer first-time freshman
-gvalen7 6.20.22*/
e.HS_SUBJ_CD = '2GRAD'

```


In addition there was an external file that was joined to this and it can be found at the file named "Highschool_SBGI_6212023.parquet" it was joined on HS_SBGI_CD = HS_CODE

Additional Highschool Details

```
select DISTINCT *  
from EDW.T_RS_ADM_STD_HS_HIST  
where campus_cd = '200'
```

This table was joined onto this table on EDW_PERS_ID

```
select DISTINCT  
EDW_PERS_ID,  
ADM_REVW_HS_GPA as hsgpa  
from EDW.V_ADM_STD_HS_HIST  
where ADM_STD_HS_CAMPUS_CD = '200'  
and ADM_STD_HS_CUR_INFO_IND = 'Y'
```

Geographic Table

```
select  
EDW_PERS_ID,  
ADDR_CITY,  
ADDR_STATE_CD,  
ADDR_ZIP_CD,  
ADDR_COUNTY_NAME,  
REG_SNAPSHOT_KEY  
from edw.t_rs_addr
```

Contains additional information about geographic location of the student.